# Assessment on LAB-03

**Even.** Design and simulate a 3-bit synchronous **UP counter** using VHDL in ModelSim. The counter should have:

- **clk** — clock input
- **reset** — active-high synchronous reset
- **enable** — counter enable input
- **Q** — **3**-bit counter output

The counter should operate according to the following conditions:

| reset | enable | Operation |
|---|---|---|
| 1 | X | Reset counter to 000 |
| 0 | 0 | Hold the current count |
| 0 | 1 | Count upward |

**Odd.**

Design and simulate a 3-bit synchronous **DOWN counter** using VHDL in ModelSim. The counter should have:

- **clk** — clock input
- **reset** — active-high synchronous reset
- **enable** — counter enable input
- **Q** — **3**-bit counter output

The counter should operate according to the following conditions:

| reset | enable | Operation |
|---|---|---|
| 1 | X | Reset counter to 111 |
| 0 | 0 | Hold the current count |
| 0 | 1 | Count downward |

# EE 4122

# Exp. : 04

**Implementation and Simulation of Logic Gates and Boolean Logic Equations Using Verilog in ModelSim**

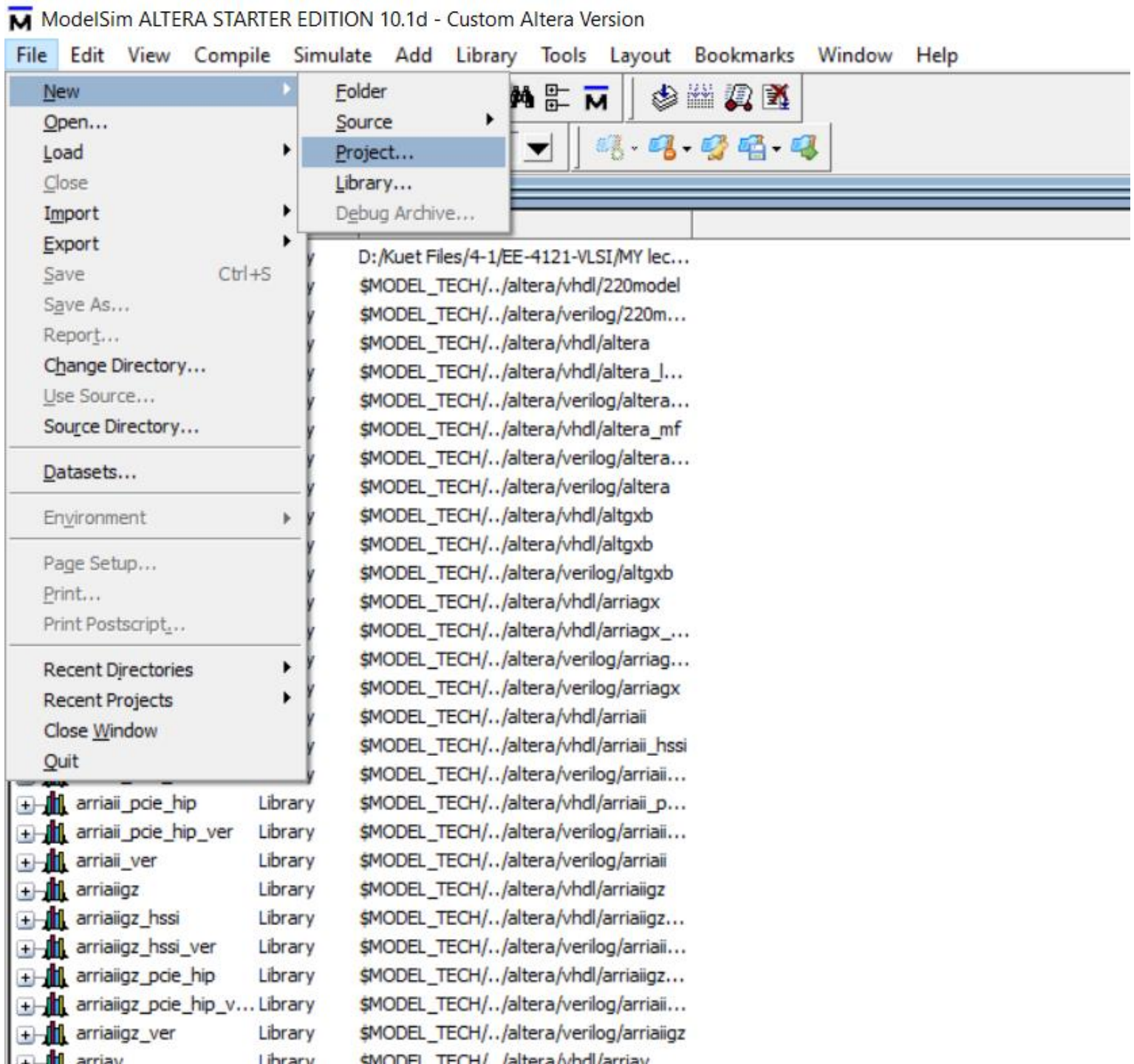
ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version

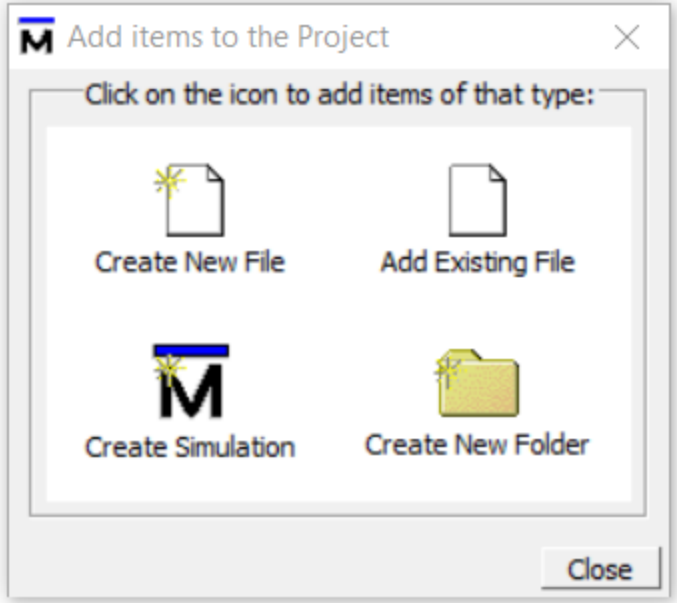
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help

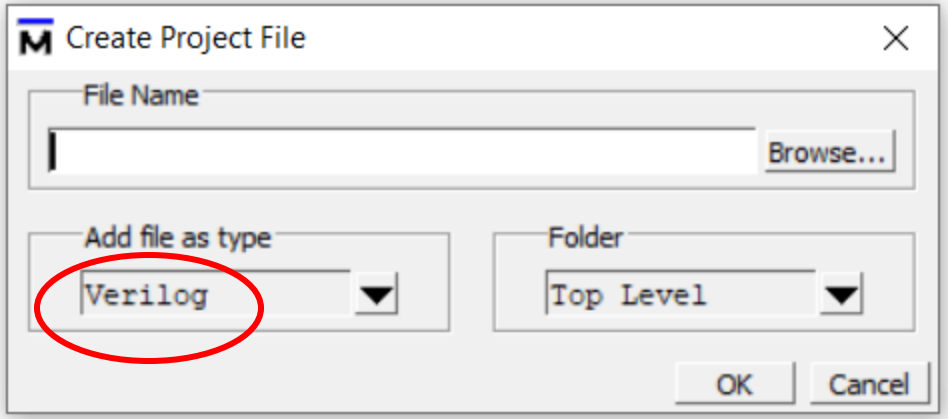
- New
- Open...
- Load
- Close Project
- Import
- Export
- Save Ctrl+S
- Save As...
- Report...
- Change Directory...
- Use Source...
- Source Directory...
- Datasets...
- Environment
- Page Setup...
- Print...
- Print Postscript...
- Recent Directories
- Recent Projects
- Close Window
- Quit

21-VLSI/MY lectures/3XORgate

| Type | Order | Modified |
| --- | --- | --- |
| VHDL | 0 | 08/07/26 08:13:12 PM |


ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Library Tools Layout Bookmarks Window Help
New
Open...
Load
Close
Import
Export
Save Ctrl+S
Save As...
Report...
Change Directory...
Use Source...
Source Directory...
Datasets...
Environment
Page Setup...
Print...
Print Postscript...
Recent Directories
Recent Projects
Close Window
Quit
Folder
Source
Project...
Library...
Debug Archive...
D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
$MODEL_TECH/../altera/vhdl/220model
$MODEL_TECH/../altera/verilog/220m...
$MODEL_TECH/../altera/vhdl/altera
$MODEL_TECH/../altera/vhdl/altera_l...
$MODEL_TECH/../altera/verilog/altera...
$MODEL_TECH/../altera/vhdl/altera_mf
$MODEL_TECH/../altera/verilog/altera...
$MODEL_TECH/../altera/verilog/altera
$MODEL_TECH/../altera/vhdl/altgxb
$MODEL_TECH/../altera/vhdl/altgxb
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$MODEL_TECH/../altera/vhdl/arriagx
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$MODEL_TECH/../altera/verilog/arriag...
$MODEL_TECH/../altera/verilog/arriagx
$MODEL_TECH/../altera/vhdl/arriaii
$MODEL_TECH/../altera/vhdl/arriaii_hssi
$MODEL_TECH/../altera/verilog/arriaii...
arriaii_pcie_hip Library
$MODEL_TECH/../altera/vhdl/arriaii_p...
arriaii_pcie_hip_ver Library
$MODEL_TECH/../altera/verilog/arriaii...
arriaii_ver Library
$MODEL_TECH/../altera/verilog/arriaii
arriaiigz Library
$MODEL_TECH/../altera/vhdl/arriaiigz
arriaiigz_hssi Library
$MODEL_TECH/../altera/vhdl/arriaiigz...
arriaiigz_hssi_ver Library
$MODEL_TECH/../altera/verilog/arriaii...
arriaiigz_pcie_hip Library
$MODEL_TECH/../altera/vhdl/arriaiigz...
arriaiigz_pcie_hip_v... Library
$MODEL_TECH/../altera/verilog/arriaii...
arriaiigz_ver Library
$MODEL_TECH/../altera/verilog/arriaiigz


Add items to the Project
Click on the icon to add items of that type:
Create New File
Add Existing File
Create Simulation
Create New Folder
Close


Create Project File
File Name
Browse...
Add file as type
Verilog
Folder
Top Level
OK
Cancel

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version

File Edit View Compile Simulate Add Source Tools Layout Bookmarks Window Help

Layout NoDesign

ColumnLayout AllColumns

Project - D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate

| Name | Status | Type | Order | Modified |
|---|---|---|---|---|
| XOR_3_gate.vhd | ? | VHDL | 0 | 08/07/26 11:27:11 PM |

D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate.vhd - Default

Ln#

1

2

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version

File Edit View Compile Simulate Add Source Tools Layout Bookmarks Window Help

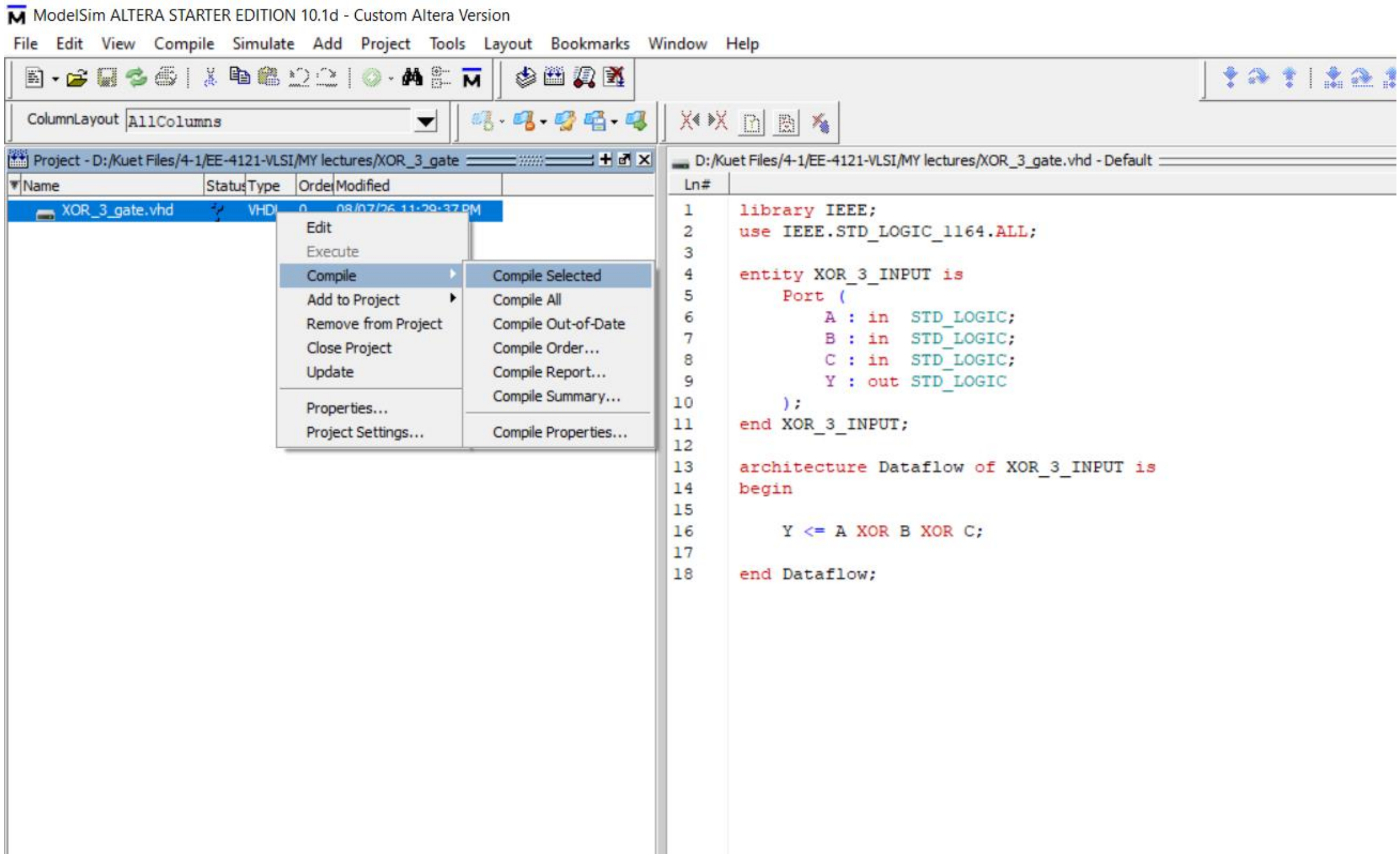
ColumnLayout AllColumns

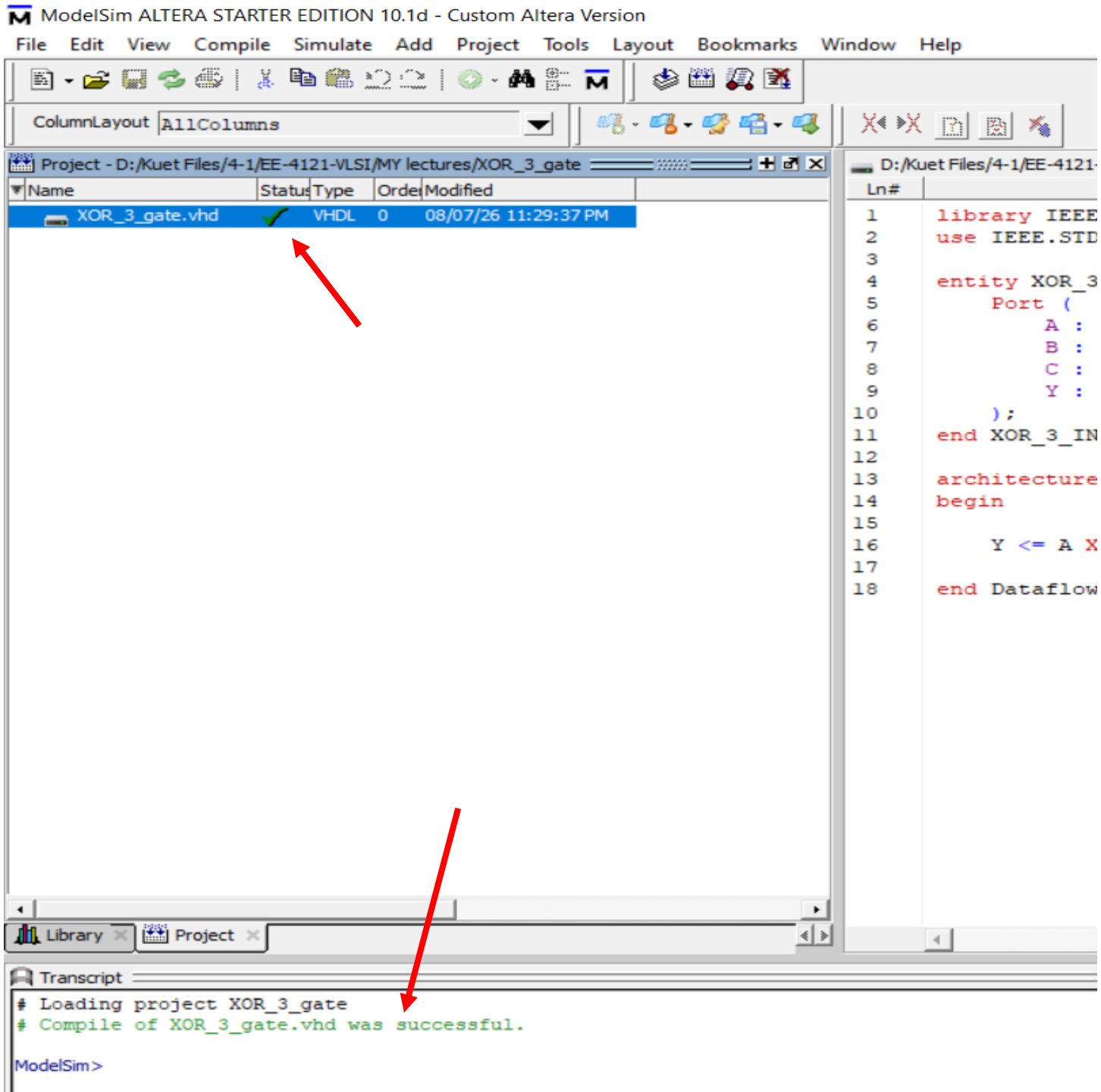
Project - D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate

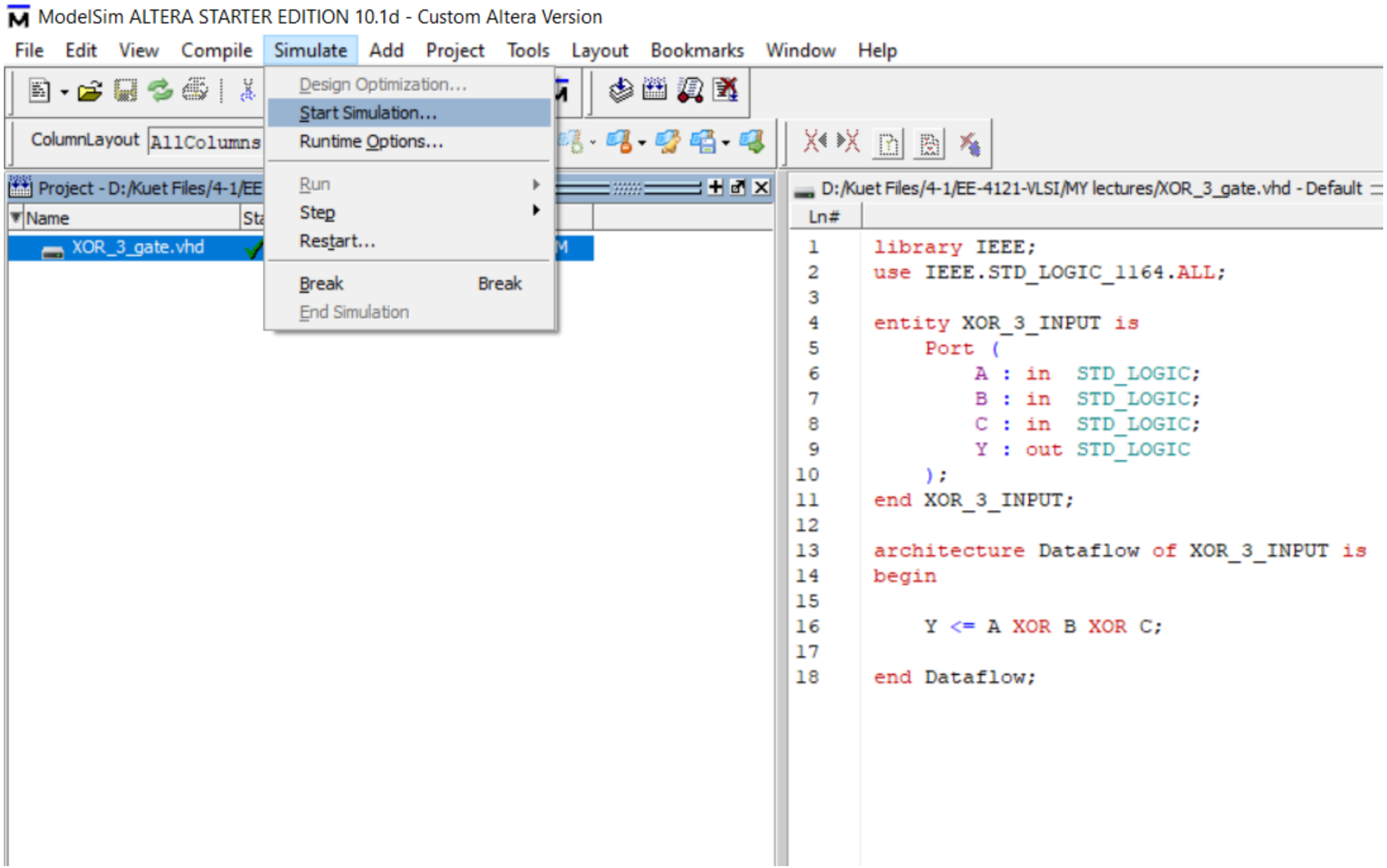
| Name | Status | Type | Orde | Modified |
|---|---|---|---|---|
| XOR_3_gate.vhd | ? | VHDL | 0 | 08/07/26 11:29:37 PM |

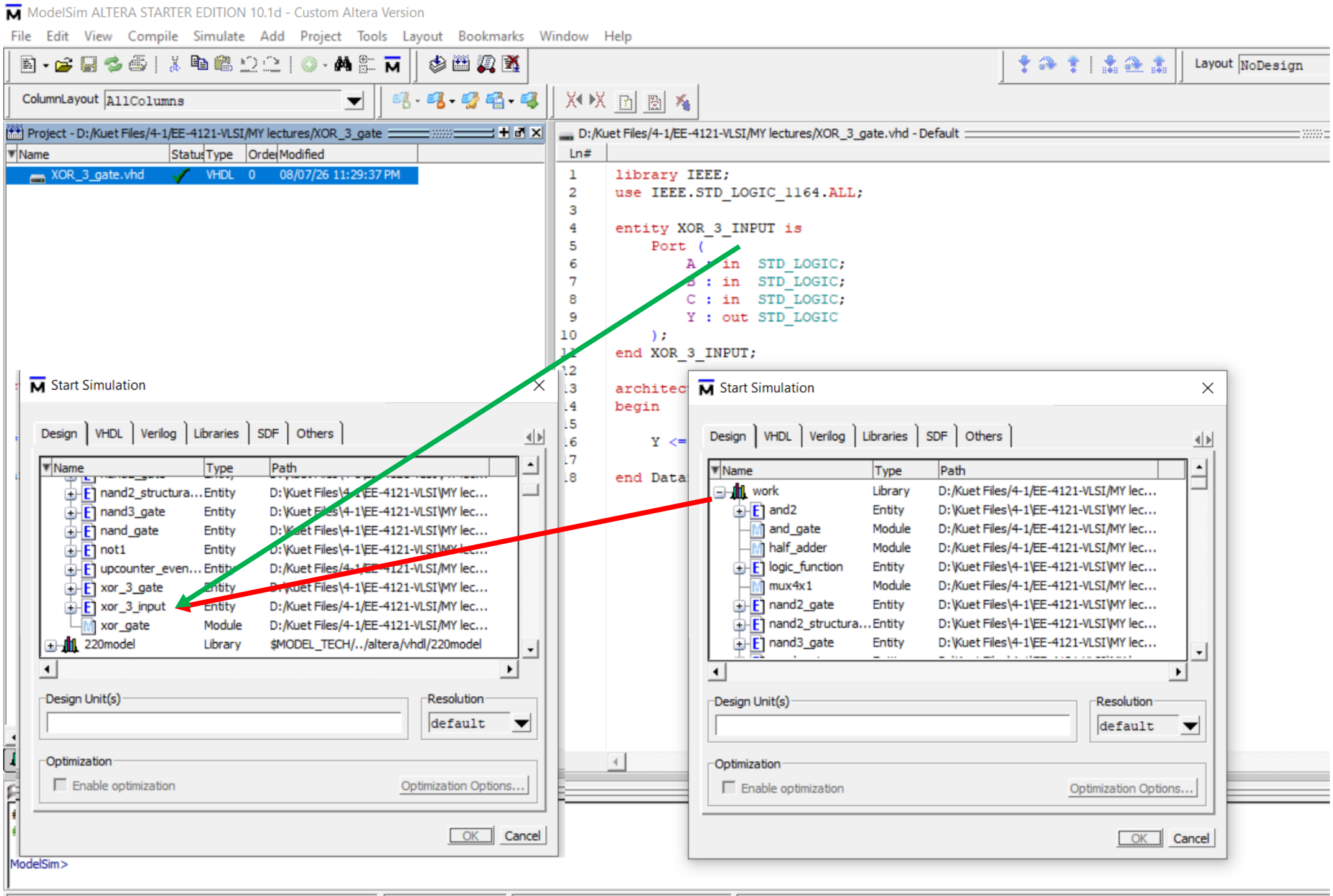
D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate.vhd - Default

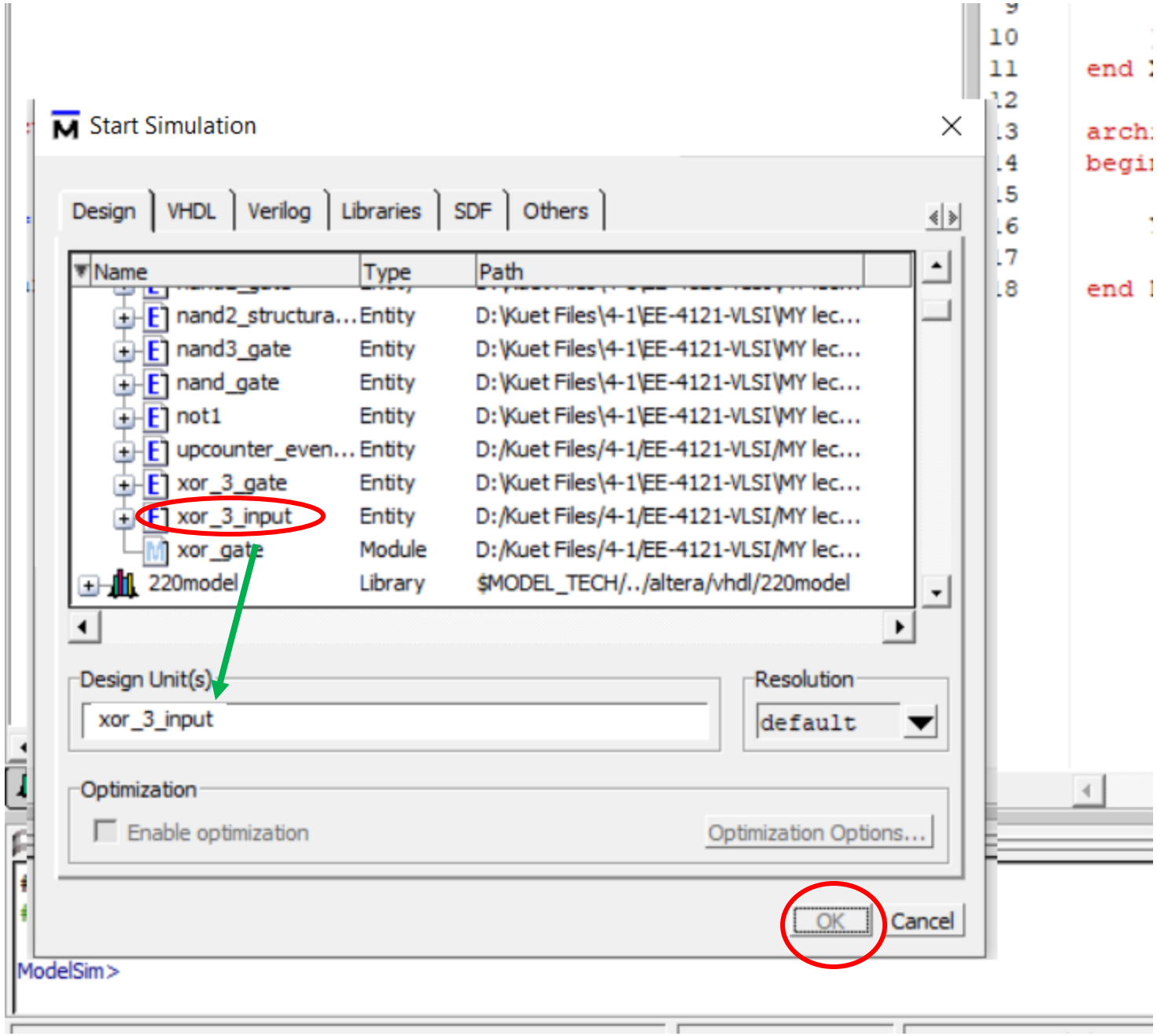
```
Ln#
 1  library IEEE;
 2  use IEEE.STD_LOGIC_1164.ALL;
 3
 4  entity XOR_3_INPUT is
 5      Port (
 6          A : in  STD_LOGIC;
 7          B : in  STD_LOGIC;
 8          C : in  STD_LOGIC;
 9          Y : out STD_LOGIC
10      );
11  end XOR_3_INPUT;
12
13  architecture Dataflow of XOR_3_INPUT is
14  begin
15
16      Y <= A XOR B XOR C;
17
18  end Dataflow;
```


ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
ColumnLayout AllColumns
Project - D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate
Name Status Type Order Modified
XOR_3_gate.vhd VHDL 0 08/07/26 11:29:37 PM
Edit
Execute
Compile
Add to Project
Remove from Project
Close Project
Update
Properties...
Project Settings...
Compile Selected
Compile All
Compile Out-of-Date
Compile Order...
Compile Report...
Compile Summary...
Compile Properties...
D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate.vhd - Default
Ln#
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3
4 entity XOR_3_INPUT is
5 Port (
6 A : in STD_LOGIC;
7 B : in STD_LOGIC;
8 C : in STD_LOGIC;
9 Y : out STD_LOGIC
10 );
11 end XOR_3_INPUT;
12
13 architecture Dataflow of XOR_3_INPUT is
14 begin
15
16 Y <= A XOR B XOR C;
17
18 end Dataflow;


ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
ColumnLayout AllColumns
Project - D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate
Name Status Type Order Modified
XOR_3_gate.vhd VHDL 0 08/07/26 11:29:37 PM
D:/Kuet Files/4-1/EE-4121-
Ln#
1 library IEEE
2 use IEEE.STD
3
4 entity XOR_3
5 Port (
6 A :
7 B :
8 C :
9 Y :
10 );
11 end XOR_3_IN
12
13 architecture
14 begin
15
16 Y <= A X
17
18 end Dataflow
Library
Project
Transcript
# Loading project XOR_3_gate
# Compile of XOR_3_gate.vhd was successful.
ModelSim>


ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Design Optimization...
Start Simulation...
Runtime Options...
Run
Step
Restart...
Break Break
End Simulation
ColumnLayout AllColumns
Project - D:/Kuet Files/4-1/EE
Name
XOR_3_gate.vhd
D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate.vhd - Default
Ln#
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3
4 entity XOR_3_INPUT is
5 Port (
6 A : in STD_LOGIC;
7 B : in STD_LOGIC;
8 C : in STD_LOGIC;
9 Y : out STD_LOGIC
10 );
11 end XOR_3_INPUT;
12
13 architecture Dataflow of XOR_3_INPUT is
14 begin
15
16 Y <= A XOR B XOR C;
17
18 end Dataflow;


ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Layout NoDesign
ColumnLayout AllColumns
Project - D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate
Name Status Type Order Modified
XOR_3_gate.vhd VHDL 0 08/07/26 11:29:37 PM
D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate.vhd - Default
Ln#
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3
4 entity XOR_3_INPUT is
5 Port (
6 A : in STD_LOGIC;
7 B : in STD_LOGIC;
8 C : in STD_LOGIC;
9 Y : out STD_LOGIC
10 );
end XOR_3_INPUT;
architec
begin
Y <=
end Data
Start Simulation
Design VHDL Verilog Libraries SDF Others
Name Type Path
nand2_structura... Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
nand3_gate Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
nand_gate Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
not1 Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
upcounter_even... Entity D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
xor_3_gate Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
xor_3_input Entity D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
xor_gate Module D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
220model Library $MODEL_TECH/../altera/vhdl/220model
Design Unit(s)
Resolution
default
Optimization
Enable optimization
Optimization Options...
OK
Cancel
Start Simulation
Design VHDL Verilog Libraries SDF Others
Name Type Path
work Library D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
and2 Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
and_gate Module D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
half_adder Module D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
logic_function Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
mux4x1 Module D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
nand2_gate Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
nand2_structura... Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
nand3_gate Entity D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
Design Unit(s)
Resolution
default
Optimization
Enable optimization
Optimization Options...
OK
Cancel
ModelSim>


Start Simulation
Design
VHDL
Verilog
Libraries
SDF
Others
Name
Type
Path
nand2_structura...
Entity
D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
nand3_gate
Entity
D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
nand_gate
Entity
D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
not1
Entity
D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
upcounter_even...
Entity
D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
xor_3_gate
Entity
D:\Kuet Files\4-1\EE-4121-VLSI\MY lec...
xor_3_input
Entity
D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
xor_gate
Module
D:/Kuet Files/4-1/EE-4121-VLSI/MY lec...
220model
Library
$MODEL_TECH/../altera/vhdl/220model
Design Unit(s)
xor_3_input
Resolution
default
Optimization
Enable optimization
Optimization Options...
OK
Cancel
ModelSim>

ModelSim ALTERA STARTER EDITION 10.1d

File Edit View Compile Simulate Add Structure Tools Layout Bookmarks Window Help

sim - Default

| Instance | Design unit | Design |
|---|---|---|
| xor_3_input | xor_3_input... | Archit |
| line__16 | xor_3_input... | Proces |
| standard | standard | Packa |
| textio | textio | Packa |
| std_logic_1164 | std_logic_1... | Packa |

Objects

| Name | Value | Kind | Mode |
|---|---|---|---|
| A | U | Signal | In |
| B | U | Signal | In |
| C | U | Signal | In |
| Y | U | Signal | Out |

Processes (Active)

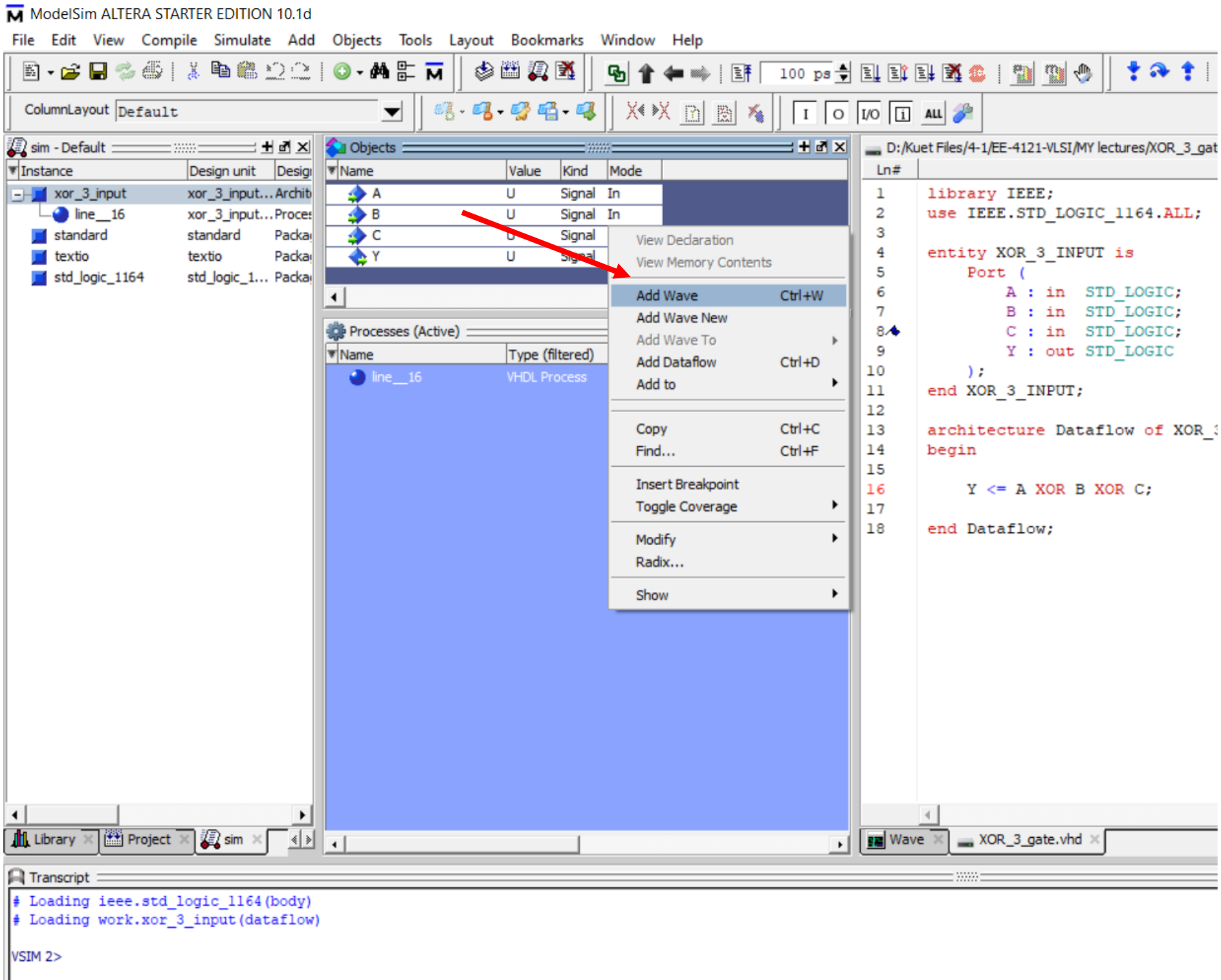
| Name | Type (filtered) | State | Order | Parent Path |
|---|---|---|---|---|
| line__16 | VHDL Process | Active | 1 | /xor_3_input |

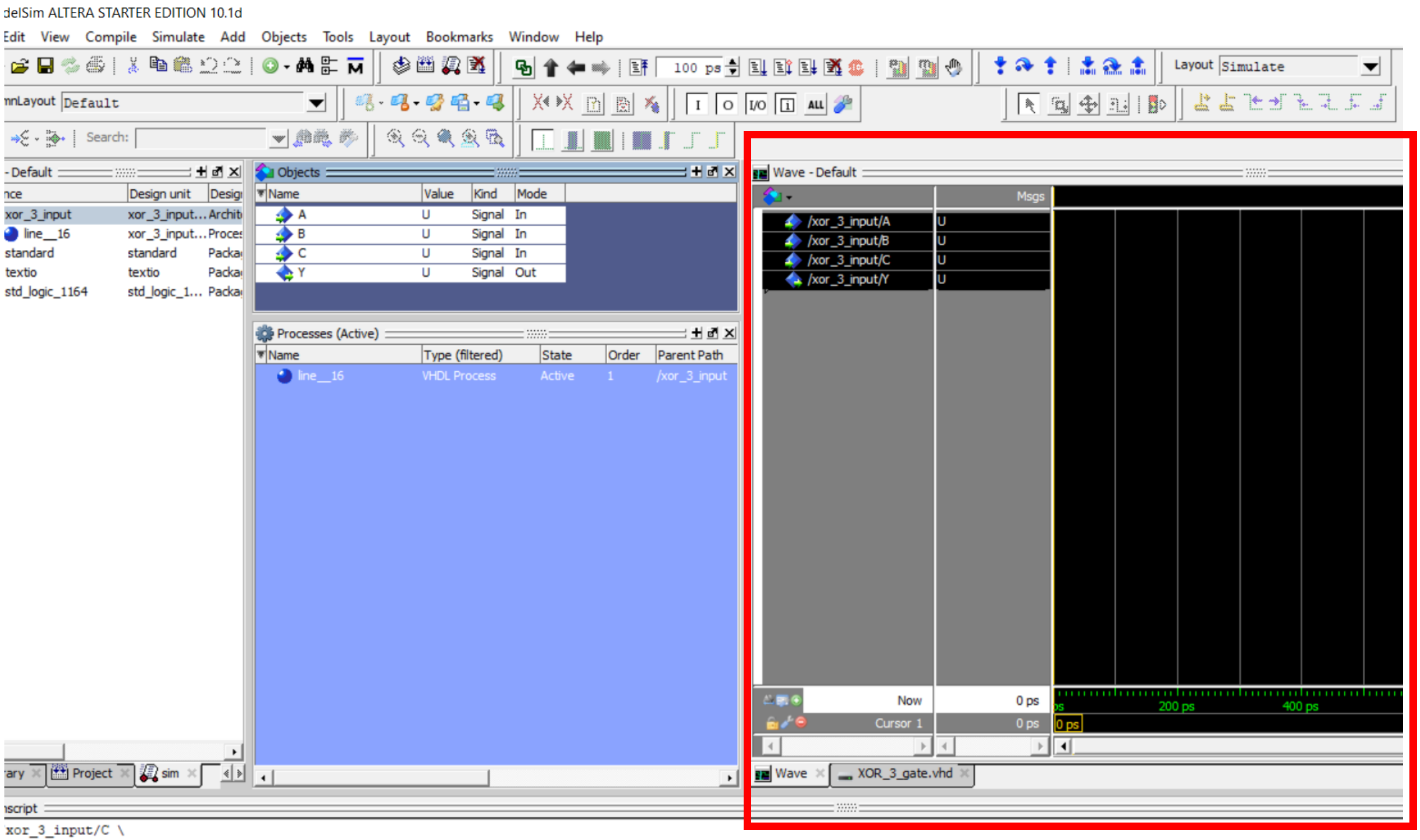
D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gate.vhd (/xor_3_input) - Default

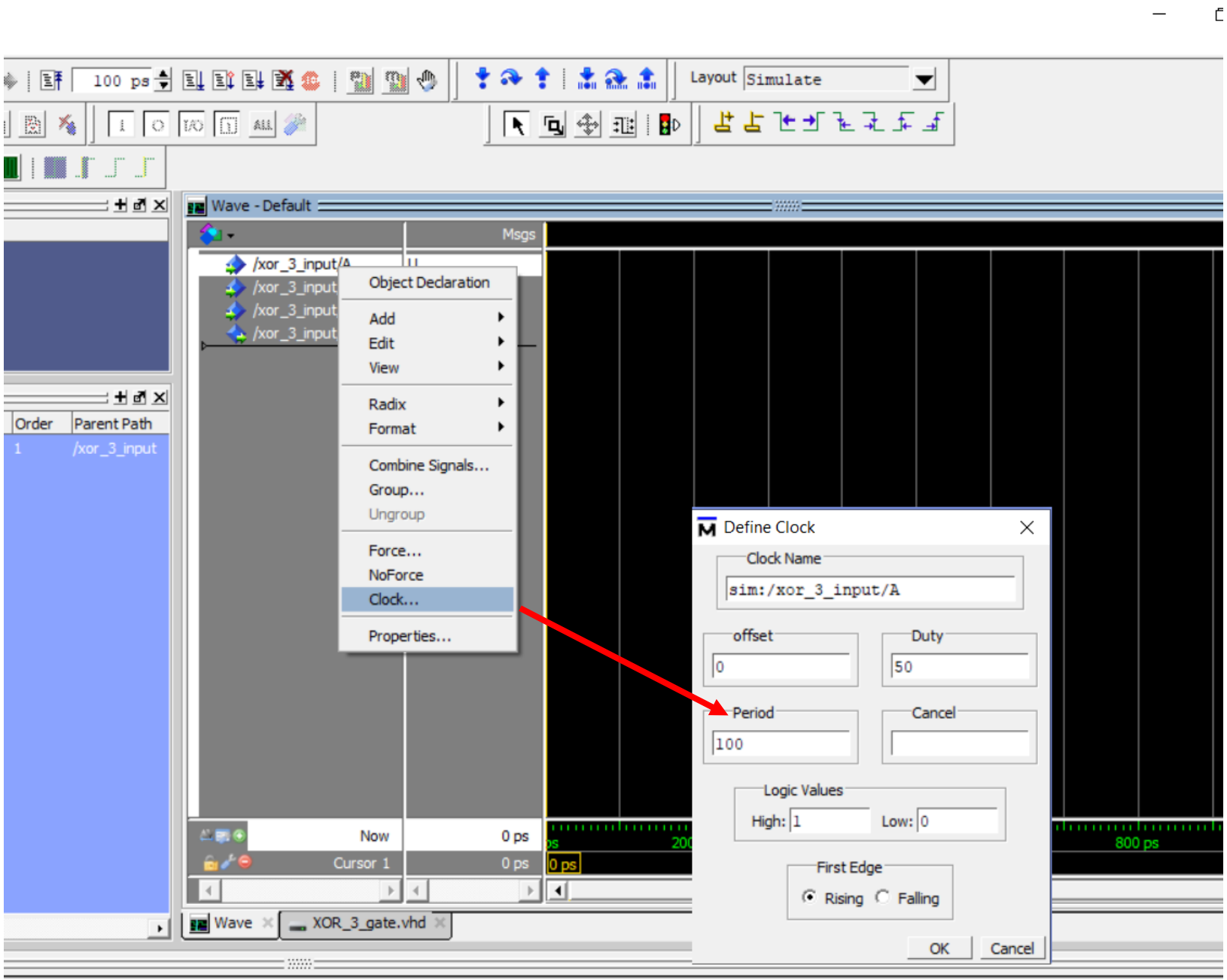
```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity XOR_3_INPUT is
5      Port (
6          A : in  STD_LOGIC;
7          B : in  STD_LOGIC;
8          C : in  STD_LOGIC;
9          Y : out STD_LOGIC
10     );
11 end XOR_3_INPUT;
12
13 architecture Dataflow of XOR_3_INPUT is
14 begin
15
16     Y <= A XOR B XOR C;
17
18 end Dataflow;
```

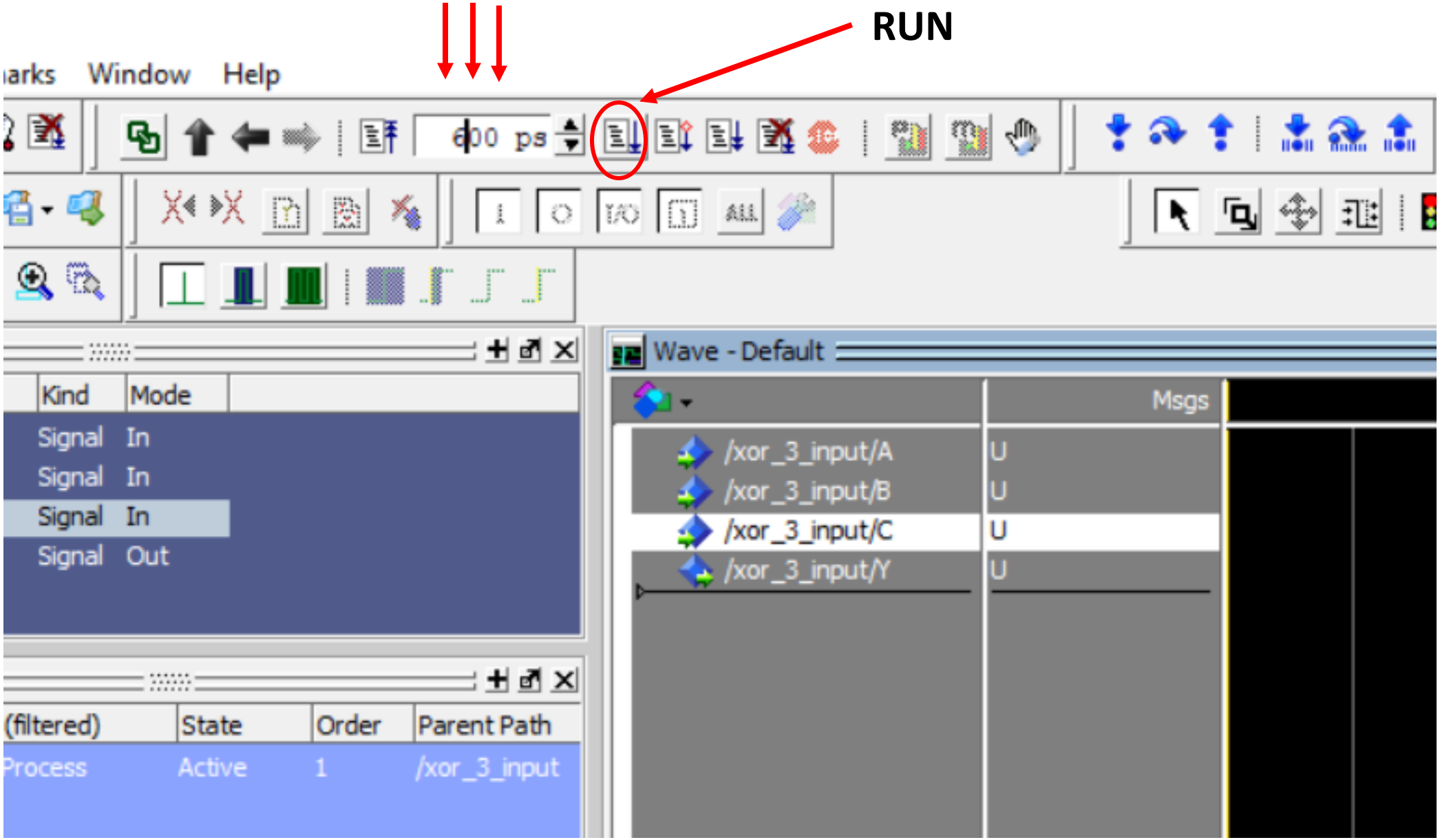
Transcript

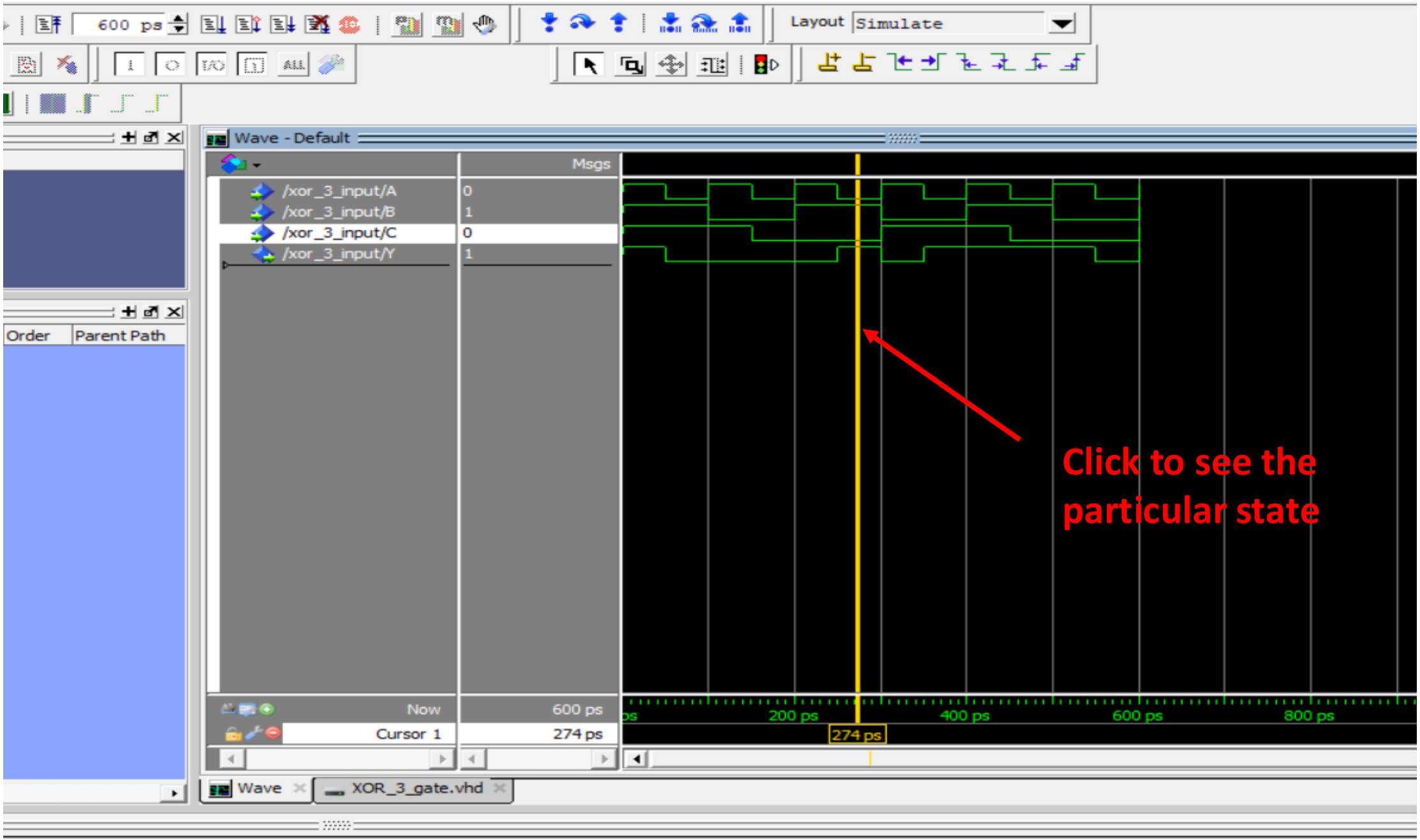
```
# Loading ieee.std_logic_1164(body)
# Loading work.xor_3_input(dataflow)
```


ModelSim ALTERA STARTER EDITION 10.1d
File Edit View Compile Simulate Add Objects Tools Layout Bookmarks Window Help
ColumnLayout Default
sim - Default
Instance Design unit
xor_3_input xor_3_input... Archit
line__16 xor_3_input... Proces
standard standard Packa
textio textio Packa
std_logic_1164 std_logic_1... Packa
Objects
Name Value Kind Mode
A U Signal In
B U Signal In
C U Signal
Y U Signal
Processes (Active)
Name Type (filtered)
line__16 VHDL Process
View Declaration
View Memory Contents
Add Wave Ctrl+W
Add Wave New
Add Wave To
Add Dataflow Ctrl+D
Add to
Copy Ctrl+C
Find... Ctrl+F
Insert Breakpoint
Toggle Coverage
Modify
Radix...
Show
D:/Kuet Files/4-1/EE-4121-VLSI/MY lectures/XOR_3_gat
Ln#
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3
4 entity XOR_3_INPUT is
5 Port (
6 A : in STD_LOGIC;
7 B : in STD_LOGIC;
8 C : in STD_LOGIC;
9 Y : out STD_LOGIC
10 );
11 end XOR_3_INPUT;
12
13 architecture Dataflow of XOR_
14 begin
15
16 Y <= A XOR B XOR C;
17
18 end Dataflow;
Library Project sim
Wave XOR_3_gate.vhd
Transcript
# Loading ieee.std_logic_1164(body)
# Loading work.xor_3_input(dataflow)
VSIM 2>


delSim ALTERA STARTER EDITION 10.1d
Edit View Compile Simulate Add Objects Tools Layout Bookmarks Window Help
100 ps
Layout Simulate
nnLayout Default
Search:
- Default
nce Design unit Desig
xor_3_input xor_3_input... Archit
line__16 xor_3_input... Proce
standard standard Packa
textio textio Packa
std_logic_1164 std_logic_1... Packa
Objects
Name Value Kind Mode
A U Signal In
B U Signal In
C U Signal In
Y U Signal Out
Processes (Active)
Name Type (filtered) State Order Parent Path
line__16 VHDL Process Active 1 /xor_3_input
Wave - Default
Msgs
/xor_3_input/A U
/xor_3_input/B U
/xor_3_input/C U
/xor_3_input/Y U
Now 0 ps
Cursor 1 0 ps
200 ps
400 ps
0 ps
Wave
XOR_3_gate.vhd
Project
sim
xor_3_input/C \
xor_3_input/Y
>


Layout
Simulate
Wave - Default
Msgs
/xor_3_input/A
/xor_3_input
/xor_3_input
/xor_3_input
Object Declaration
Add
Edit
View
Radix
Format
Combine Signals...
Group...
Ungroup
Force...
NoForce
Clock...
Properties...
Order
Parent Path
1
/xor_3_input
Now
0 ps
Cursor 1
0 ps
0 ps
800 ps
Wave
XOR_3_gate.vhd
Define Clock
Clock Name
sim:/xor_3_input/A
offset
0
Duty
50
Period
100
Cancel
Logic Values
High: 1
Low: 0
First Edge
Rising
Falling
OK
Cancel


RUN
arks Window Help
Kind Mode
Signal In
Signal In
Signal In
Signal Out
(filtered) State Order Parent Path
Process Active 1 /xor_3_input
Wave - Default
Msgs
/xor_3_input/A U
/xor_3_input/B U
/xor_3_input/C U
/xor_3_input/Y U


600 ps
Layout
Simulate
Wave - Default
Msgs
/xor_3_input/A
/xor_3_input/B
/xor_3_input/C
/xor_3_input/Y
0
1
0
1
Order
Parent Path
Now
600 ps
Cursor 1
274 ps
200 ps
400 ps
600 ps
800 ps
274 ps
Click to see the particular state
Wave
XOR_3_gate.vhd

## 4 bit up-counter##

```
module up_counter_4bit (
    input clk,
    input reset,
    input enable,
    output reg [3:0] Q
);

always @(posedge clk) begin
    if (reset)
        Q <= 4'b0000;
    else if (enable)
        Q <= Q + 4'b0001;
    else
        Q <= Q;
end

endmodule
```

## 3 bit down counter##

```
module down_counter (
    input       clk,
    input       reset,
    input       enable,
    output reg [2:0] Q
);

always @(posedge clk) begin
if (reset)
            Q <= 3'b111;
else if (enable)
            Q <= Q - 3'b001;
else
            Q <= Q;
    end

endmodule
```